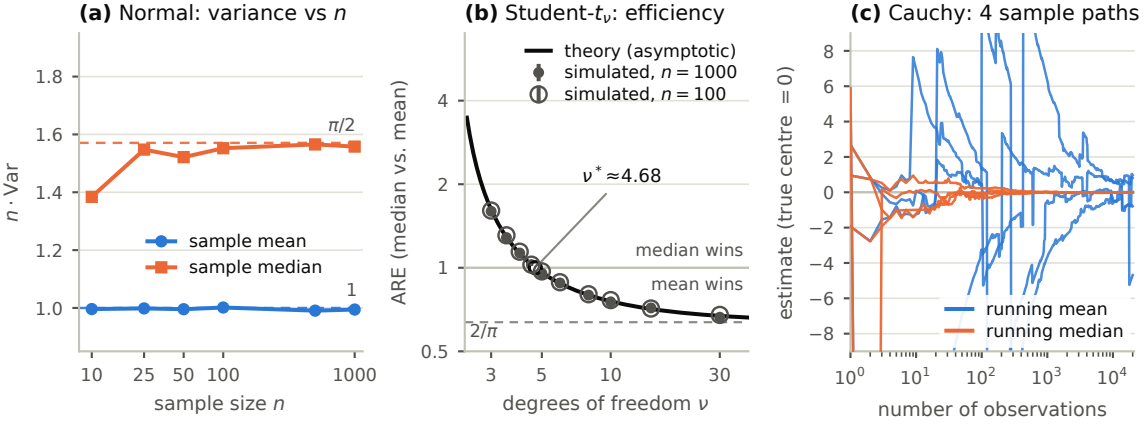
(a) Normal: variance vs n
n · Var
1.8
1.6
1.4
1.2
1.0
π/2
1
sample mean
sample median
10
25
50
100
1000
sample size n
(b) Student-t_ν: efficiency
ARE (median vs. mean)
4
2
1
0.5
theory (asymptotic)
simulated, n = 1000
simulated, n = 100
ν* ≈ 4.68
median wins
mean wins
2/π
3
5
10
30
degrees of freedom ν
(c) Cauchy: 4 sample paths
estimate (true centre = 0)
8
6
4
2
0
−2
−4
−6
−8
running mean
running median
10^0
10^1
10^2
10^3
10^4
number of observations